

BUILD YOUR PERSONAL CYBERSECURITY LAB

Task-2 Report

Submitted By

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Objective

The objective of this task is to create a safe and isolated cybersecurity practice environment using virtualization technology. This lab environment helps in learning penetration testing, vulnerability assessment, traffic analysis, and web application security testing without affecting real-world systems.

Tools and Software Used

Tool	Purpose
VMware Workstation	Virtualization platform
Kali Linux	Attacker machine
Docker	Container platform
OWASP Juice Shop	Vulnerable web application
Nmap	Network scanning
Wireshark	Network traffic analysis
Burp Suite	Web security testing

System Requirements

- RAM: Minimum 8GB
- Disk Space: 40GB free
- Operating System: Windows 11
- Internet Connection

Lab Architecture

The cybersecurity lab consists of:

- Host Operating System (Windows 11)
- Kali Linux Virtual Machine
- NAT Network Adapter
- Host-Only Network Adapter
- OWASP Juice Shop vulnerable application running inside Kali Linux

This setup provides an isolated environment for safe cybersecurity practice.

Step-by-Step Implementation

Step 1: Installing VMware Workstation

VMware Workstation was already installed on the host system. It was used to create and manage the Kali Linux virtual machine.

Step 2: Setting Up Kali Linux

The Kali Linux VMware image was imported into VMware Workstation and powered on successfully.

Network Configuration

Two network adapters were configured:

- Adapter 1 → NAT
- Adapter 2 → Host-Only

This allowed internet connectivity and isolated lab communication.

Step 3: Updating Kali Linux

The system repositories were updated and package issues were fixed using Linux package management commands.

Commands used:

```
sudo apt update  
sudo apt upgrade -y
```

Step 4: Installing Docker

Docker was installed successfully to deploy the vulnerable web application.

Command used:

```
sudo apt install docker.io -y
```

Docker service was started using:

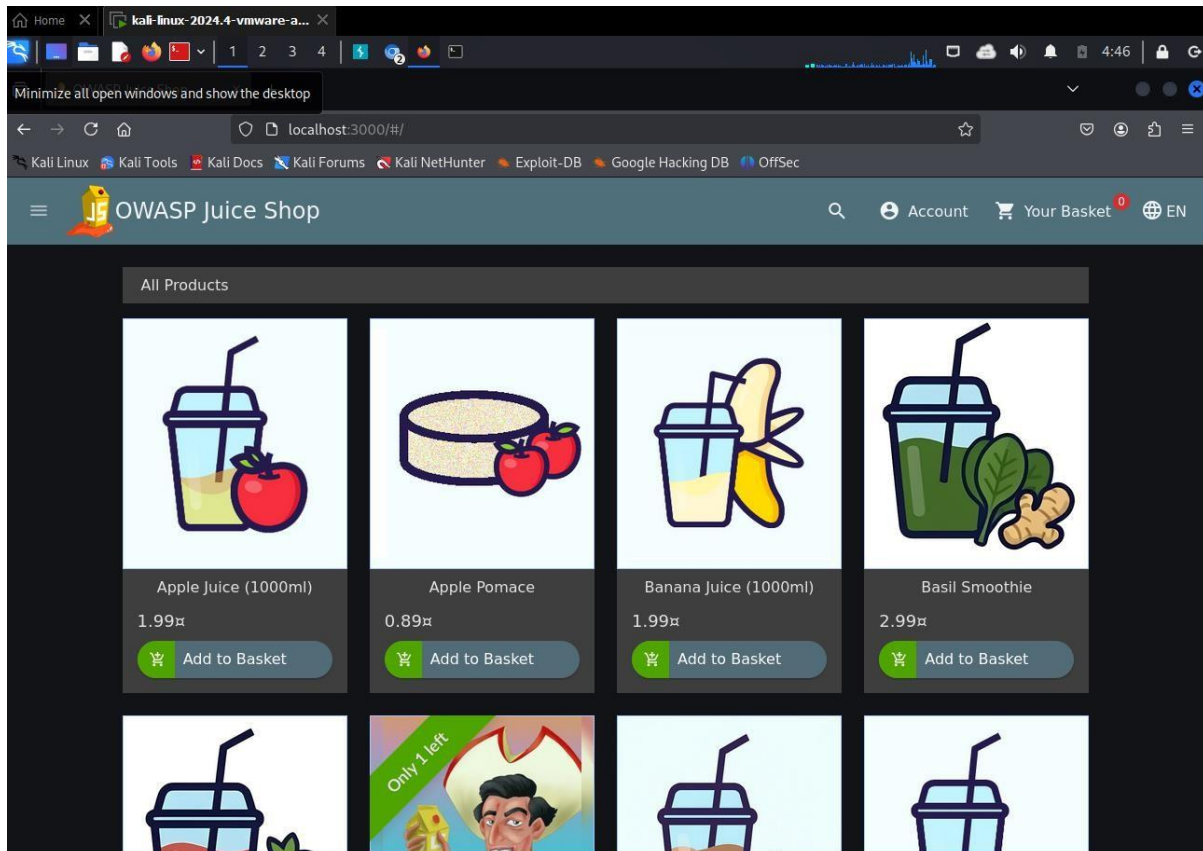
```
sudo systemctl start docker
```



```
(kali@kali)-[~]
$ sudo docker run -d -p 3000:3000 bkimminich/juice-shop
Unable to find image 'bkimminich/juice-shop:latest' locally
latest: Pulling from bkimminich/juice-shop
fa8ae93e2b3a: Pull complete
c172f21841df: Pull complete
b4e6f1bfce0a: Pull complete
b4242723c53f: Pull complete
d6b1b89eccac: Pull complete
2780920e5dbf: Pull complete
7c12895b777b: Pull complete
3214acf345c0: Pull complete
52630fc75a18: Pull complete
dd64bf2dd177: Pull complete
b839d9ae01f6: Pull complete
ebddc55facdc: Pull complete
bdfd7f7e5bf6: Pull complete
4ec36cb7292a: Pull complete
b73b1097e3a1: Pull complete
bd8962e29291: Pull complete
cac2ae0193cb: Pull complete
dd0d9bfd3bee: Pull complete
7cc70dfd88cf: Pull complete
8224d91da70b: Pull complete
1a4be5562d92: Pull complete
7281cbae89a9: Pull complete
7b7c492cc60b: Pull complete
fe1d31fae6c6: Pull complete
Digest: sha256:25fd268112350ae9e0ddc7878371f9f12f5b0b546c7bf934d6599aa8e724418f
Status: Downloaded newer image for bkimminich/juice-shop:latest
2d36924bae76f8e42a07a8f2c2cf00458c0298dac214e1b80fd1f6588db360c7
```

The application was accessed through:

<http://localhost:3000>



Step 6: Network Scanning using Nmap

Nmap was used to scan open ports and identify services running on the target system.

Command used:

```
nmap -sV localhost
```

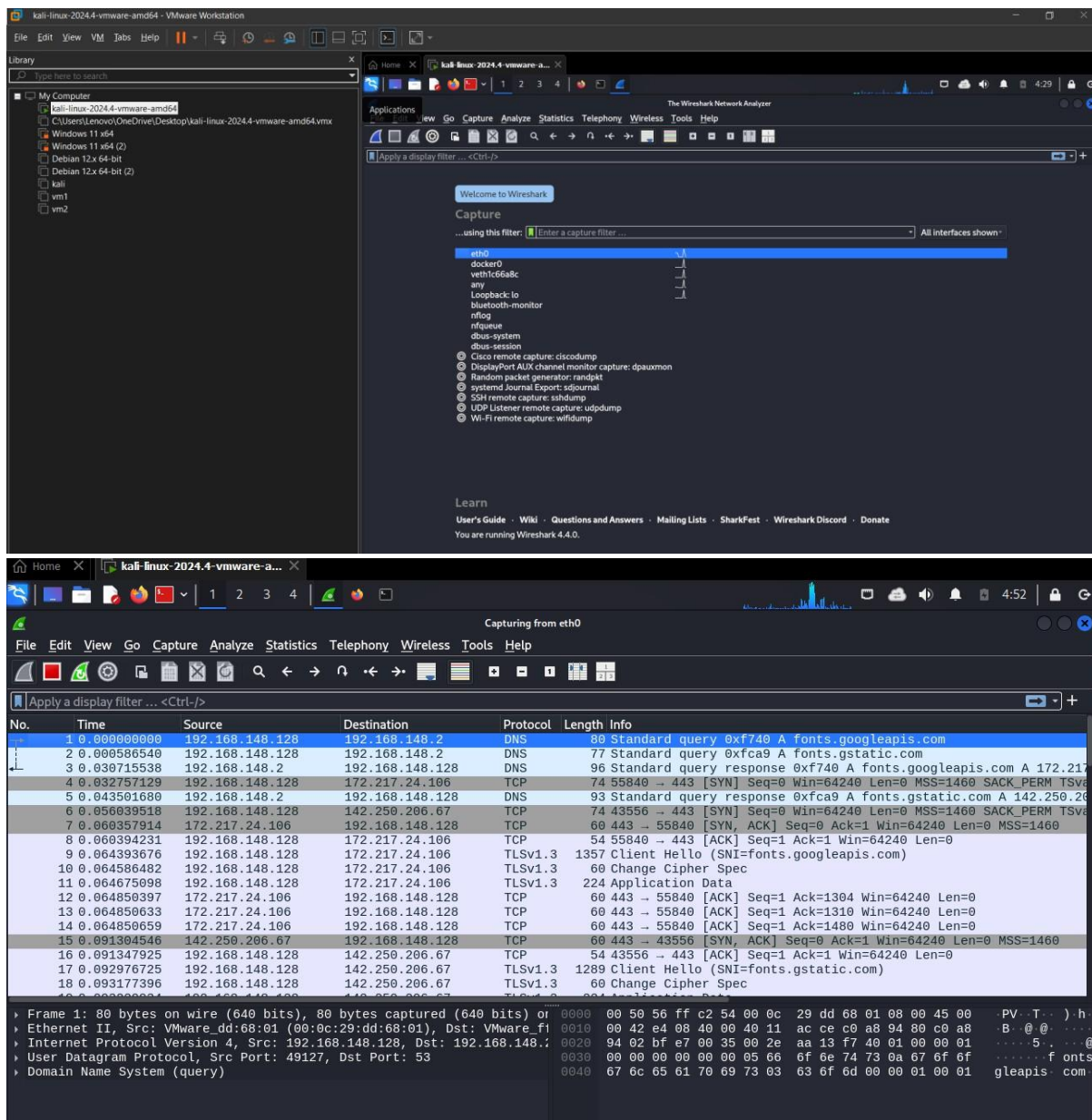
```
(kali@kali)~$ nmap -v localhost
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-05-26 04:26 EDT
Nmap scan report for localhost (127.0.0.1)
Host is up (0.0000040s latency).
Other addresses for localhost (not scanned): ::1
Not shown: 999 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
3000/tcp   open  ppp?
1 service unrecognized despite returning data. If you know the service/version, please submit the following fingerprint at https://nmap.org/cgi-bin/submit.cgi?inow=service:
SF-ProT3000-TCP-V=7.94SVNRKI-7X0-5/26xTime=6A15595AXP-x86_64-pc-linux-gnuXr
SF:(GetRequest,2866,"HTTP/1.\x20200\x200K\r\nAccess-Control-Allow-Origin
SF:::\x20*\r\nX-Content-Type-Options:\x20nosniff\r\nX-Frame-Options:\x20SA
SF:MEORIGIN\r\nFeature-Policy:\x20payment\x20'self'\r\nX-Recruiting:\x20;
SF:/jobs\r\nAccept-Ranges:\x20bytes\r\nCache-Control:\x20public,\x20max-ag
SF:=0\r\nLast-Modified:\x20Tue,\x202026\x20May\x202026T\x2008:14:44\x20GMT\r
SF:nEndTimeTag:\x20M-\x26af-19e6359b56c'\r\nContent-Type:\x20text/html;\x20cha
SF:rsset=UTF-8'\r\nContent-Length:\x20209938\r\nVary:\x20Accept-Encoding\r\nDoS
SF:Size:\x20True\x20May\x202026T\x2008:27:06\x20GMT\r\nConnection:\x20c
SF:lose\r\n\r\n#-----\n\n\x20\x20\x20Copyright\x20(c)\x202014-2026\x20Bjoer
SF:n\x20Kimminich\x206\x20the\x20OWASP\x20Juice\x20Shop\x20Contributors,\n
SF:n\x20\x20\x20\x20SPDX-License-Identifier:\x20MIT\n\n\x20\x20-->\n\n<!doctype
SF:y>\x20html>\n<html\x20lang=en'\x20data-beasties-container>\nhead>\n\n\x
SF:20\x20meta\x20charset=utf-8'\x20\n\x20\x20titleOWASP\x20Juice\x20Sho
SF:p</title>\n\n\x20\x20meta\x20name=description'\x20content=\x20Probably
SF:tothe\x20most\x20modern\x20and\x20sophisticated\x20insecure\x20web\x20
SF:application'\x20\n\x20\x20meta\x20name=viewport'\x20content=width&
SF:vice-width,\x20initial-scale=1'\x20\n\x20link\x20rel=pconnect'\x20refo
SF:r=http'\x20helpof HTTP/1.1\x202000\x20Bad\x20request'\nConne
SF:ction:\x20close\r\n\r\n}\x20(MCP,2F,"HTTP/1.1\x202040\x20Bad\x20Reque
SF:nConnection:\x20close\r\n\r\n")\x20(HTTPOptions,EA,"HTTP/1.1\x20204\x2
SF:NO\x20Content\r\nAccess-Control-Allow-Origin:\x20*\r\nAccess-Control-
SF:Allow-Methods:\x20GET,HEAD,PUT,PATCH,POST,DELETE\r\nVary:\x20Access-Con
SF:trol-Request-Headers\r\nContent-Length:\x2020\r\nDate:\x20Tue,\x202026\x20
SF:May\x202026T\x2008:27:06\x20GMT\r\nConnection:\x20close\r\n\r\n")\x20(RTSP
SF:Request,EA,"HTTP/1.1\x20204\x20No\x20Content\r\nAccess-Control-Allow-O
SF:igin:\x20*\r\nAccess-Control-Allow-Methods:\x20GET,HEAD,PUT,PATCH,POS
SF:T,DELETE\r\nVary:\x20Access-Control-Request-Headers\r\nContent-Length:\x
SF:200\r\nDate:\x20Tue,\x202026\x20May\x202026T\x2008:27:06\x20GMT\r\nConnec
SF:tion:\x20close\r\n\r\n");
```

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 11.42 seconds

The scan successfully identified running services.

Step 7: Traffic Analysis using Wireshark

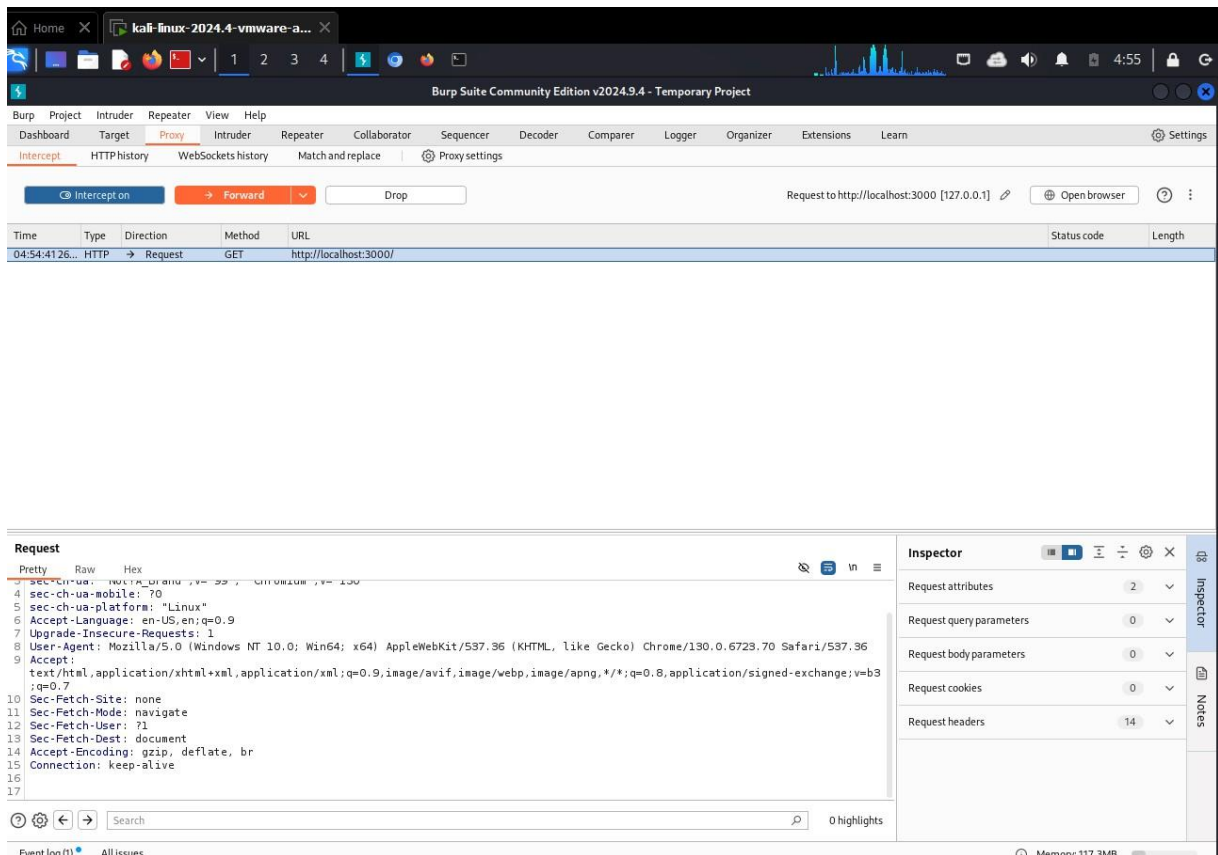
Wireshark was used to capture and analyze network packets generated while accessing OWASP Juice Shop.



Packet capturing was successfully performed on the network interface.

Step 8: Web Application Testing using Burp Suite

Burp Suite Community Edition was used to intercept HTTP requests between the browser and OWASP Juice Shop.



The proxy interception feature successfully captured web traffic requests.

Results

The cybersecurity lab environment was successfully created using VMware and Kali Linux. OWASP Juice Shop was deployed successfully using Docker. Network scanning, traffic analysis, and web request interception were performed successfully using Nmap, Wireshark, and Burp Suite respectively.

Screenshots Included

1. VMware Workstation setup
2. Kali Linux virtual machine
3. IP configuration using `ip a`
4. Docker container running
5. OWASP Juice Shop webpage

6. Nmap scanning results
7. Wireshark packet capture
8. Burp Suite interception

Conclusion

Successfully created a personal cybersecurity lab environment using virtualization technology. The lab provided hands-on experience with network scanning, packet analysis, and web application security testing tools in a safe and isolated environment.

References

- [Kali Linux Documentation](#)
- [OWASP Juice Shop Documentation](#)
- [Docker Documentation](#)
- [Nmap Official Documentation](#)
- [Wireshark Official Documentation](#)
- [Burp Suite Community Edition Documentation](#)